ARTSPACE / CAPSTONE PROJECT

ArtSpace

Valentina Bernal Moreno

Software Engineer

LinkedIn

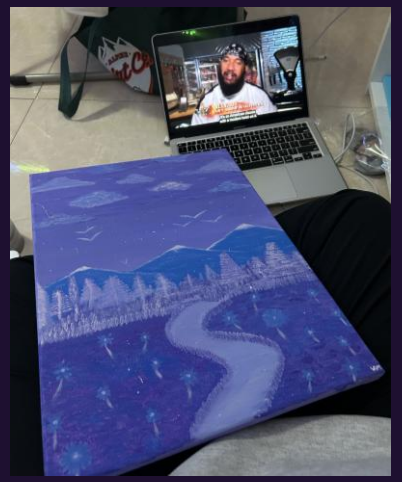
Tuesday October 13th 2026

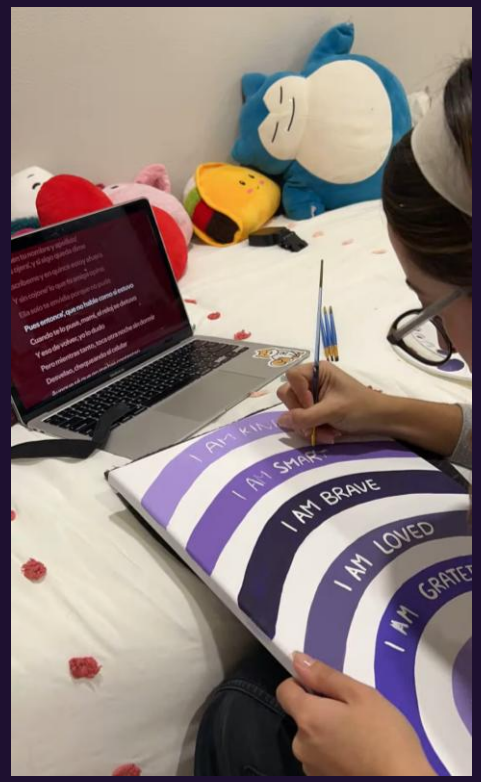
ABOUT ME

- I was born in Bogotá, Colombia. I am 23 years old
- Web Development – Brewster Technical College
- Part of the Software Engineering Apprenticeship.
- Customer service and leadership roles across the fast-food and banking industries
- Communication, teamwork, adaptability, and problem-solving skills.

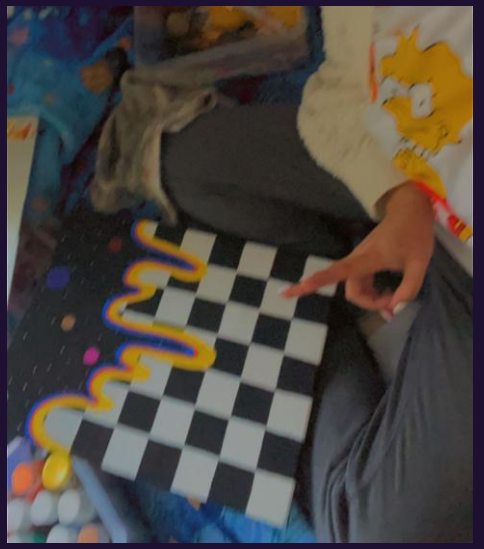
WHERE ART MEETS CODE!!

Outside of technology, one of my favorite creative outlets is art, especially painting, which became the inspiration behind my capstone project, ArtSpace.









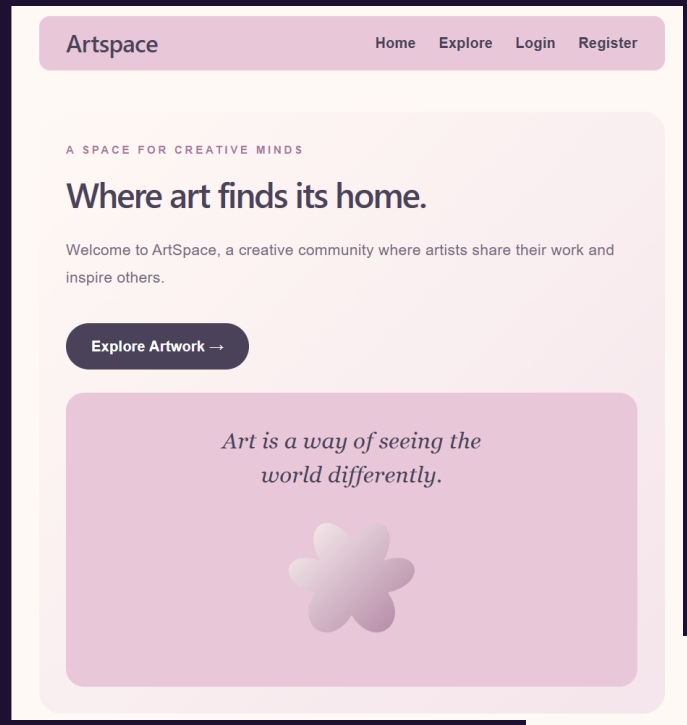
ArtSpace

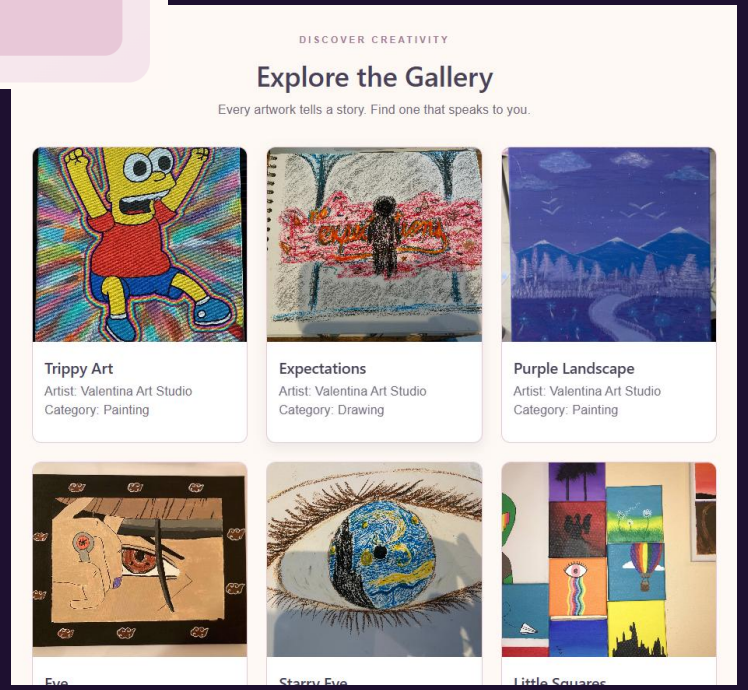
WHAT IS ARTSPACE?

An online gallery that connects artists with future artists that are looking for inspiration.

THE MAIN FEATURE

Users upload an image, add details, and publish their work to the online gallery.





1 / Problem & solution

Why build an online gallery?



THE PROBLEM

Artists can struggle finding inspiration to create art.

THE SOLUTION

ArtSpace provides a personal gallery: upload an image, title, description and category.

Visitors can browse; authenticated artists manage their work.

One upload creates a portfolio item that others can discover.

2 / Architecture & upload flow

How requests flow from the browser through the frontend and backend to the database

CONTROLLER

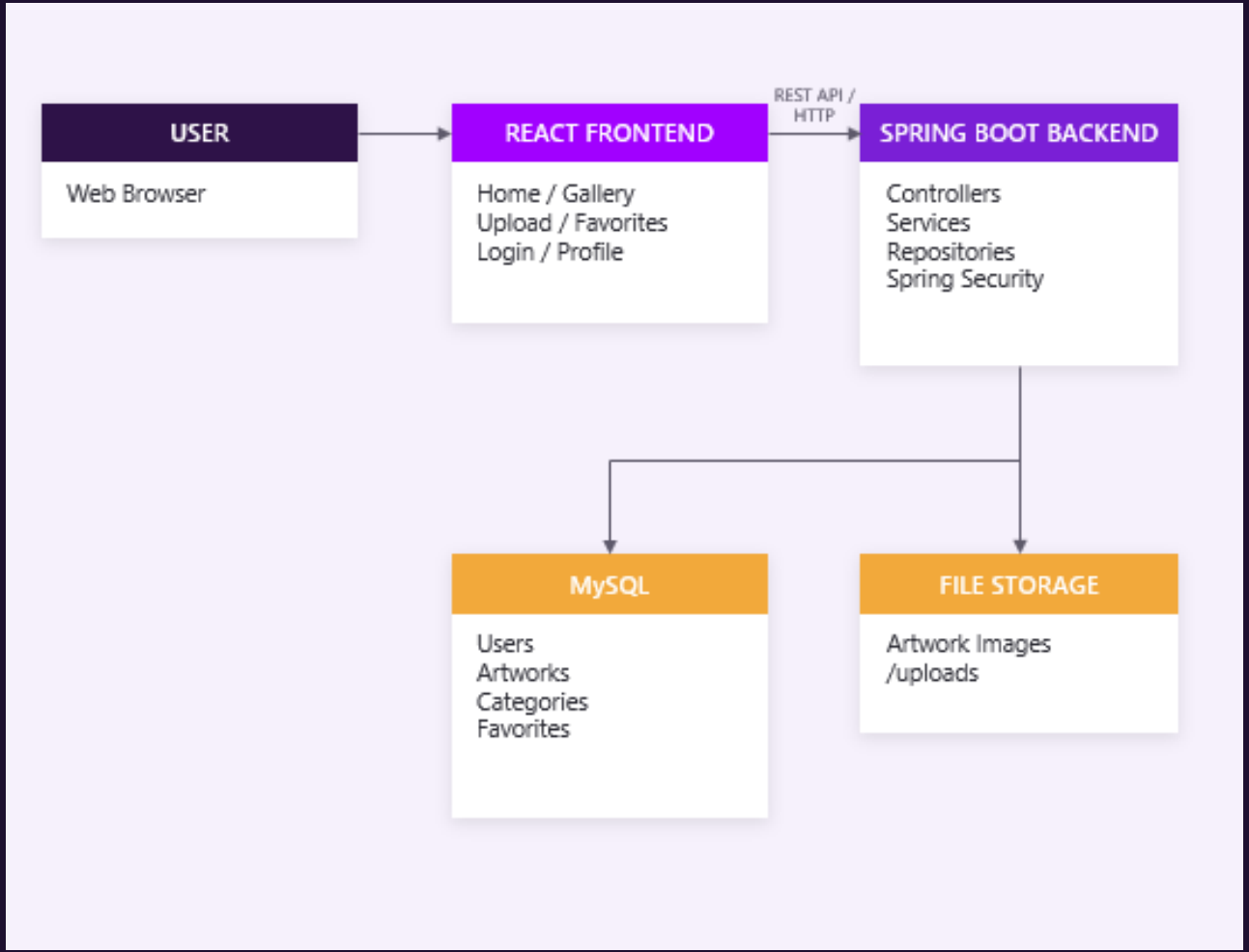
Accepts the multipart HTTP request

FILE STORAGE

In charge of saving and managing the actual artwork images that users upload

SPRING SECURITY

Controls access to protected uploads



MYSQL + /uploads

Stores artwork

= Title, artist, category, image path

REPOSITORIES

Responsible for communicating with our database

SERVICES

Validates and applies business rules

3 / Entity • Define an artwork

```java
@Entity  Valentina BM
@Table(name = "artworks")
public class Artwork {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(nullable = false)  3 usages
    private LocalDateTime createdAt;

    @Column(nullable = false, length = 150)  3 usages
    private String title;

    @Column(length = 1000)  3 usages
    private String description;

    @Column(nullable = false)  3 usages
    private String imagePath;

    @ManyToOne(optional = false)  3 usages
    @JoinColumn(name = "user_id", nullable = false)
    private User user;

    @ManyToOne(optional = false)  3 usages
    @JoinColumn(name = "category_id", nullable = false)
    private Category category;
```

KEY FIELDS

description: artwork details

imagePath: location of image file

createdAt: timestamp

user_id: artist relationship

category_id: category relationship

@ManyToOne + @JoinColumn map each artwork to one user and one category.

4 / Repository • Talk to MySQL

ArtworkRepository.java → Spring Data JPA

```
7
8   public interface ArtworkRepository extends JpaRepository<Artwork, Long> {  8 usages  Valentina BM
9
10      List<Artwork> findByUserId(Long userId);  3 usages  Valentina BM
11
12      List<Artwork> findByCategoryId(Long categoryId);  3 usages  Valentina BM
13  }
```

WHAT IT PROVIDES

JpaRepository supplies save(), findAll() and findById(). Custom methods filter by user ID or category ID.

5 / Service • Validate before saving

ArtworkService.create() → business rules

```
43
44 @    public Artwork create(Artwork input) {  Valentina BM
45          if (input.getUser() == null || input.getCategory() == null) {
46              return null;
47          }
48          User user = userRepository.findById(input.getUser().getId()).orElse( other: null);
49          if (user == null) {
50              return null;
51          }
52          Category category = categoryRepository.findById(input.getCategory().getId()).orElse( other: null);
53          if (category == null) {
54              return null;
55          }
56          if (!fileStorageService.isValidImagePath(input.getImagePath())) {
57              return null;
58          }
59          input.setUser(user);
60          input.setCategory(category);
61          return artworkRepository.save(input);
62      }
```

VALIDATION

Check user and category are provided.

Look up both records.

Verify image path.

Only then call repository.save().

6 / Controller • Receive the upload

ArtworkController.createWithImage() → REST API

```
56
57      @PostMapping  Valentina BM
58  @   public ResponseEntity<Artwork> create(@RequestBody Artwork input, Principal principal) {
59          User user = userService.findByEmail(principal.getName());
60          input.setUser(user);
61          Artwork saved = artworkService.create(input);
62          if (saved == null) {
63              return ResponseEntity.badRequest().build();
64          }
65          return ResponseEntity.status(201).body(saved);
66      }
```

HTTP ENDPOINT

This method receives an Artwork object through @RequestBody.

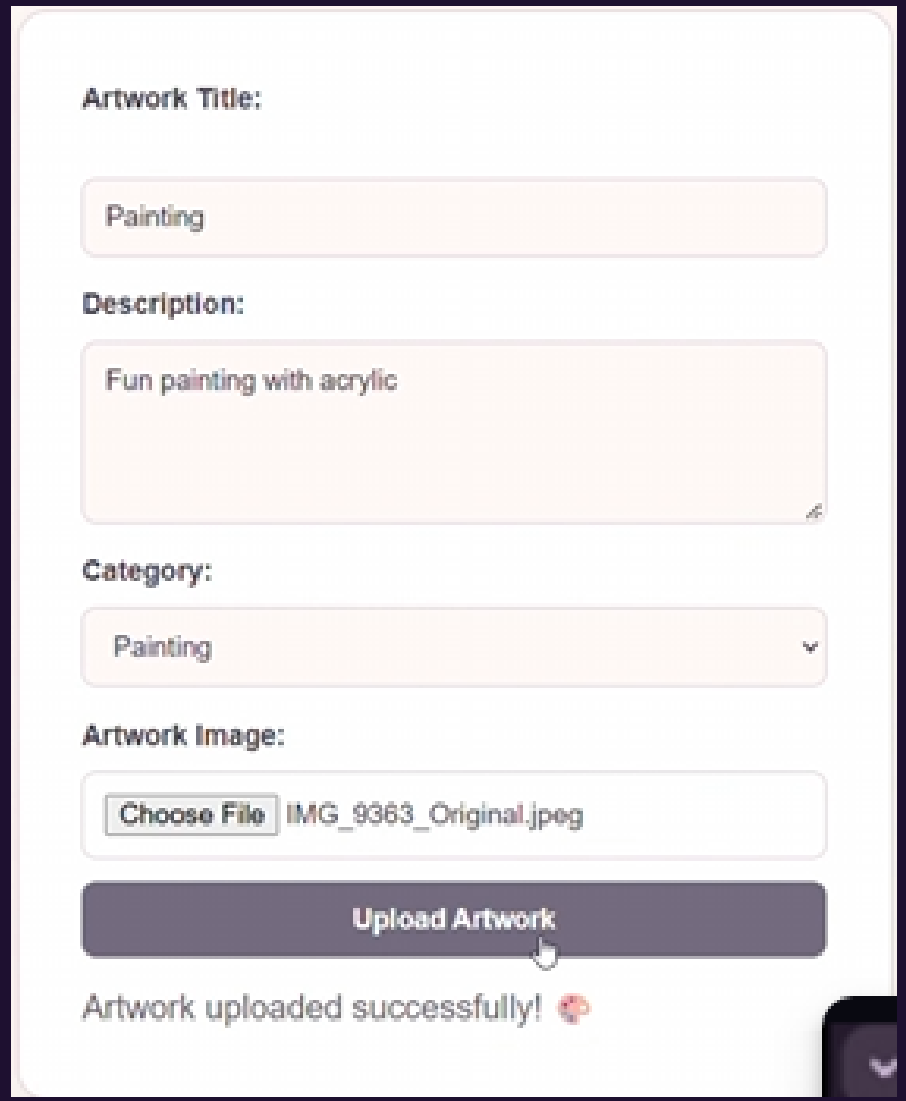
@RequestBody Artwork input converts incoming JSON into an Artwork object.

userService.findByEmail(...)

Retrieves the logged-in artist’s

account.input.setUser(user)Associates the artwork with its creator.

7 / Lessons learned



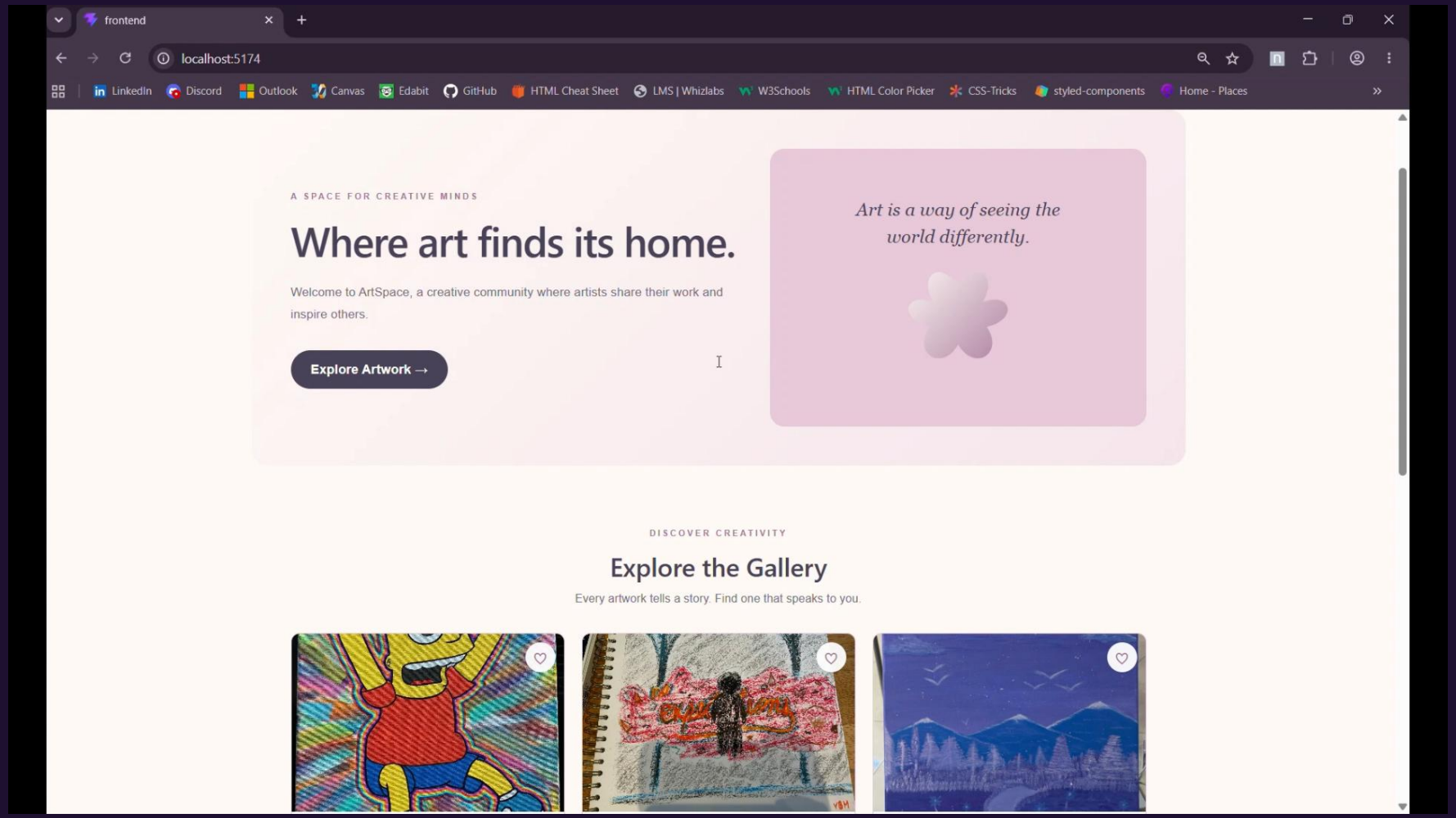
CHALLENGE → SOLUTION

- Bringing My Vision to Life = Balancing my attention to detail with the time available to develop ArtSpace.
- Finding the Right API = Choosing Pixabay to support artistic exploration and inspiration.
- Balancing Design and Functionality = Creating a visually appealing application while ensuring the frontend and backend worked together.

KEY TAKEAWAY

- I learned to balance creativity, problem-solving, and technical development to bring my vision to life.

8 / Live demo


frontend
localhost:5174
LinkedIn
Discord
Outlook
Canvas
Edabit
GitHub
HTML Cheat Sheet
LMS | Whizlabs
W3Schools
HTML Color Picker
CSS-Tricks
styled-components
Home - Places
A SPACE FOR CREATIVE MINDS
Where art finds its home.
Welcome to ArtSpace, a creative community where artists share their work and inspire others.
Explore Artwork →
Art is a way of seeing the world differently.
DISCOVER CREATIVITY
Explore the Gallery
Every artwork tells a story. Find one that speaks to you.

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Thank you!

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